

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: T Y B Tech Branch : Chemical Engineering Semester : VI

Subject Code & Name: (BTCHO605A) Pharmaceuticals & Fine Chemical

Max Marks: 60

Date: 30/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	Impurities in pharmaceutical preparation may be due to following sources:	2/CO2	1				
	<table border="1"> <tr> <td>a. Raw material</td> <td>b. Manufacturing process</td> <td>c. Chemical instability</td> <td>d. All of the above</td> </tr> </table>	a. Raw material	b. Manufacturing process	c. Chemical instability	d. All of the above		
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2	.In Gas-liquid phase chromatography, the stationary phase is composed of _____ or the mobile stage is made of _____	4/C3	1				
	<table border="1"> <tr> <td>a. Solid, liquid</td> <td>b. Liquid, liquid</td> <td>c. Liquid, gas</td> <td>d. Solid, gas</td> </tr> </table>	a. Solid, liquid	b. Liquid, liquid	c. Liquid, gas	d. Solid, gas		
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3	The gravimetric analysis particularly uses	4/CO2	1				
	<table border="1"> <tr> <td>a. precipitation and drying of residue</td> <td>b precipitation only</td> <td>C dissolution</td> <td>d.complexation</td> </tr> </table>	a. precipitation and drying of residue	b precipitation only	C dissolution	d.complexation		
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4	Pharmaceutical buffer system could be categorizes into .	1/CO2	1				
	<table border="1"> <tr> <td>a. 1</td> <td>b. 2</td> <td>c. 3</td> <td>d. 4</td> </tr> </table>	a. 1	b. 2	c. 3	d. 4		
a. 1	b. 2	c. 3	d. 4				
5	Drugs converted to suitable form are known as... b) c)	2/CO3	1				
	<table border="1"> <tr> <td>a. Excipient</td> <td>b. Source of drug</td> <td>c. Dosage form</td> <td>d. API</td> </tr> </table>	a. Excipient	b. Source of drug	c. Dosage form	d. API		
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6	Which powder must be sterilized before their use.....				3/CO5	1
	a. Medicated dusting powders	b. Non medicated powders	c Natural dusting powder	d. Surgical dusting powders		
7	Which category of drugs is evaluated for dissolution?				3/CO3	1
	a. coated tablets	b .solutions	c. suspension	d. uncoated tablets		
8	The most common granulation method is				3/CO4	1
	a. Dry granulation	b. Wet granulation	c. Direct compression	d. All of the above		
9	Which test is most satisfactory for determining the drug content uniformity of the tablets				4/CO4	1
	a. Dissolution	b. disintegration	c. Weight variation	d. Friability		
10	One of the parameter is not affected by compression force in tableting process.				4/CO4	1
	a. Hardness	b. Density	c. Disintegration	d. Surface area		
11	The points of interest for the development of a successful capsule dosage form are				5/CO4	1
	a. Physical and chemical stability	b. Fill composition	C Shell material	d. All of the above		
12	Which of the following is not a method of dry heat sterilization				2/CO4	1
	a. Flaming	b . Hot air oven	c. Incineration	d. Pasteurization		
Q. 2	Solve Any Two the following.					12
A)	Define the term impurity. Explain various sources of impurities present in pharmaceuticals. Explain the Principle for the Limit test for sulphate				2/CO1	6
B)	Discuss the method of preparation, uses and assay of sodium				2/CO2	6

	chloride and calcium Gluconate.		
C)	Explain the solvents used in non-aqueous titrations. Explain the solvents used in non-aqueous titrations.	2/CO2	
Q.3	Solve Any Two the following.		12
A)	Describe limit test of iron as per IP and give role of iron free citric acid and iron free ammonia in the test	2/CO2	6
B)	Define quality control and give its importance in pharmacy.	1/CO2	6
C)	Define and classify dosage forms with suitable examples.	1/CO3	
Q. 4	Solve Any Two of the following.		12
A)	Write process description of Metformine with neat process flow diagram.	2/CO3	6
B)	Explain the role of Active Pharmaceutical Ingredients in pharmaceutical products.	2/CO4	6
C)	Explain the process of formation of amoxicillin along with flow diagram.	2/CO4	6
Q.5	Solve Any Two of the following.		12
A)	Enumerate various tests for quality control of tablets. Describe the disintegration or friability test. Give any four reasons for film coating.	1/CO4	6
B)	Advantages of tablet dosage form over other oral drug delivery systems	4/CO4	6
C)	Discuss the tablet Ingredients/ Excipients with their functions	2/CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Define capsule. Describe the process involved in manufacturing of hard gelatin capsule. List different excipients used in processing of capsule.	2/CO5	6
B)	Define sterilization. Classify different methods used for sterilization.	1/CO5	6
C)	Differentiate between sterilization and disinfection. Explain the process of sterilization using radiation.	4/CO5	6
	*** End ***		